

0.1 Construction of the Real Numbers: Completeness Axiom, Dedekind Cuts, and Cauchy Sequences

The fundamental inadequacy of the rational number field Q lies in its lack of completeness; it contains gaps, most famously demonstrated by the absence of a rational root for the equation $x^2 = 2$. To establish a rigorous foundation for analysis, we must construct the field of real numbers R such that it satisfies the Least Upper Bound property. We examine two classical constructions: Dedekind cuts and equivalence classes of Cauchy sequences, ultimately proving their equivalence in yielding a complete ordered field.

0.1.1 Dedekind Cuts

A **Dedekind cut** is a subset $\alpha \subset Q$ satisfying the following three properties:

1. **Nontriviality:** $\alpha \neq \emptyset$ and $\alpha \neq Q$.
2. **Downward Closure:** If $p \in \alpha$, $q \in Q$, and $q < p$, then $q \in \alpha$.
3. **No Greatest Element:** If $p \in \alpha$, there exists $r \in \alpha$ such that $p < r$.

Let R denote the set of all Dedekind cuts. We define an ordering on R by strict set inclusion: $\alpha < \beta$ if and only if $\alpha \subset \beta$.

[Completeness Axiom for Cuts] The set R of Dedekind cuts satisfies the Least Upper Bound property. That is, every non-empty subset $A \subset R$ that is bounded above possesses a supremum in R .

Let A be a non-empty subset of R that is bounded above. This implies there exists some cut $\beta \in R$ such that $\alpha \subset \beta$ for all $\alpha \in A$. We define the candidate supremum γ as the union of all cuts in A :

$$\gamma = \bigcup_{\alpha \in A} \alpha$$

We must first verify that γ is a valid Dedekind cut.

1. Since A is non-empty, there exists some $\alpha_0 \in A$. Because α_0 is a cut, $\alpha_0 \neq \emptyset$, hence $\gamma \neq \emptyset$. Furthermore, since every $\alpha \in A$ is bounded above by β , $\gamma \subset \beta$. As $\beta \neq Q$, it follows that $\gamma \neq Q$.
2. Let $p \in \gamma$ and $q \in Q$ with $q < p$. By definition of γ , $p \in \alpha$ for some $\alpha \in A$. Since α is downward closed, $q \in \alpha$, which implies $q \in \gamma$.
3. Let $p \in \gamma$. Then $p \in \alpha$ for some $\alpha \in A$. Since α has no greatest element, there exists $r \in \alpha$ such that $p < r$. Thus, $r \in \gamma$, demonstrating that γ has no greatest element.

Therefore, $\gamma \in R$. By construction, for any $\alpha \in A$, $\alpha \subset \gamma$, so γ is an upper bound for A . Suppose $\delta < \gamma$ is another upper bound for A . Then there exists $p \in \gamma$ such that $p \notin \delta$. Since $p \in \gamma$, $p \in \alpha$ for some $\alpha \in A$. But this implies $\alpha \not\subset \delta$, contradicting the assumption that δ is an upper bound for A . Thus, γ is the least upper bound of A .

0.1.2 Cauchy Sequences and the Completion of Q

An alternative, topologically motivated construction of R relies on Cauchy sequences.

A sequence $(q_n)_{n=1}^\infty$ in Q is a **Cauchy sequence** if for every rational $\epsilon > 0$, there exists an integer $N \in \mathbb{N}$ such that for all $m, n \geq N$,

$$|q_m - q_n| < \epsilon$$

Let $\mathcal{C}$ denote the set of all Cauchy sequences in Q . We define an equivalence relation $\sim$ on $\mathcal{C}$ by declaring $(q_n) \sim (p_n)$ if and only if $\lim_{n \rightarrow \infty} |q_n - p_n| = 0$.

The relation $\sim$ is an equivalence relation on $\mathcal{C}$.

Reflexivity and symmetry are immediate from the properties of the absolute value function. For transitivity, suppose $(q_n) \sim (p_n)$ and $(p_n) \sim (r_n)$. Let $\epsilon > 0$ be given. There exist $N_1, N_2 \in \mathbb{N}$ such that $|q_n - p_n| < \frac{\epsilon}{2}$ for all $n \geq N_1$ and $|p_n - r_n| < \frac{\epsilon}{2}$ for all $n \geq N_2$. Let $N = \max(N_1, N_2)$. For $n \geq N$, the triangle inequality yields:

$$|q_n - r_n| \leq |q_n - p_n| + |p_n - r_n| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Hence, $\lim_{n \rightarrow \infty} |q_n - r_n| = 0$, proving $(q_n) \sim (r_n)$.

We construct R as the quotient space $\mathcal{C}/\sim$. An element $x \in R$ is thus an equivalence class of Cauchy sequences of rational numbers.

The field R constructed via Cauchy sequences is complete. That is, every Cauchy sequence in R converges to a limit in R .

Let $(x_n)_{n=1}^\infty$ be a Cauchy sequence in R . Each x_n is an equivalence class of rational Cauchy sequences. For each $n \in \mathbb{N}$, choose a representative rational sequence $(q_{n,k})_{k=1}^\infty \in x_n$. Since x_n represents a real number, there exists $K_n \in \mathbb{N}$ such that for all $j, k \geq K_n$,

$$|q_{n,j} - q_{n,k}| < \frac{1}{n}$$

We extract a diagonal sequence $y_n = q_{n,K_n}$. We claim that $(y_n)_{n=1}^\infty$ is a Cauchy sequence in Q , and its equivalence class $y \in R$ is the limit of the sequence (x_n) .

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that $\frac{3}{N} < \epsilon$. Since (x_n) is Cauchy in R , there exists $M \geq N$ such that for all $m, n \geq M$, $|x_m - x_n| < \frac{\epsilon}{3}$. By the definition of the real metric, this implies that for sufficiently large k , $|q_{m,k} - q_{n,k}| < \frac{\epsilon}{3}$.

Consider $m, n \geq M$. We have:

$$|y_m - y_n| = |q_{m,K_m} - q_{n,K_n}| \leq |q_{m,K_m} - q_{m,k}| + |q_{m,k} - q_{n,k}| + |q_{n,k} - q_{n,K_n}|$$

For $k \geq \max(K_m, K_n)$ large enough to satisfy the inter-sequence bound, this yields:

$$|y_m - y_n| < \frac{1}{m} + \frac{\epsilon}{3} + \frac{1}{n} \leq \frac{1}{M} + \frac{\epsilon}{3} + \frac{1}{M} \leq \frac{2}{N} + \frac{\epsilon}{3} < \epsilon$$

Thus, (y_n) is a Cauchy sequence in Q . Let $y = [(y_n)]$. A similar bounding argument demonstrates that $|x_n - y| \rightarrow 0$ as $n \rightarrow \infty$, proving that R is complete under the metric induced by this construction.